



No Code Visualization Tools Power BI

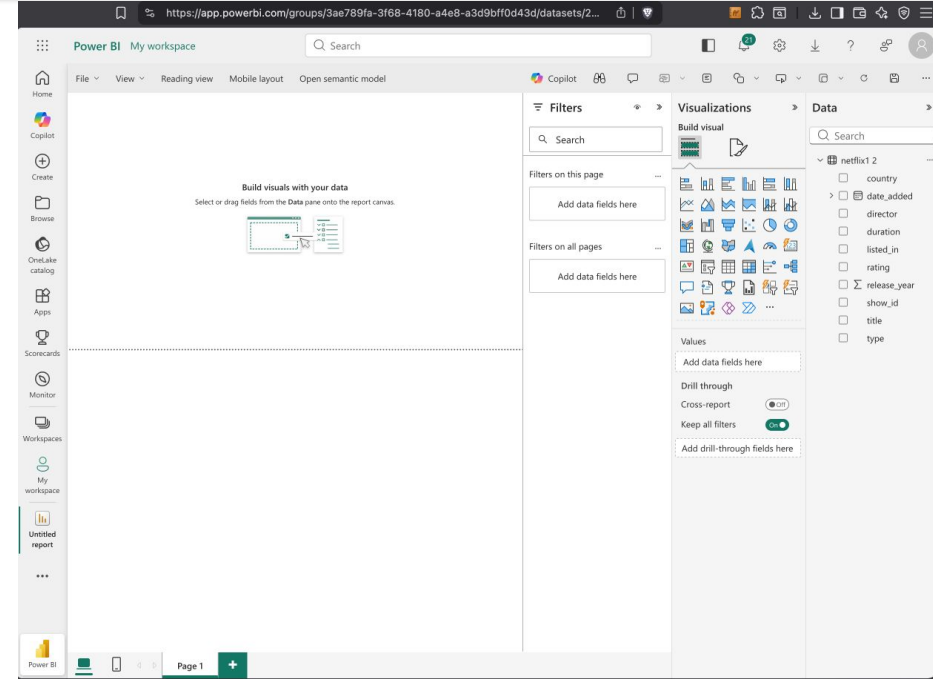
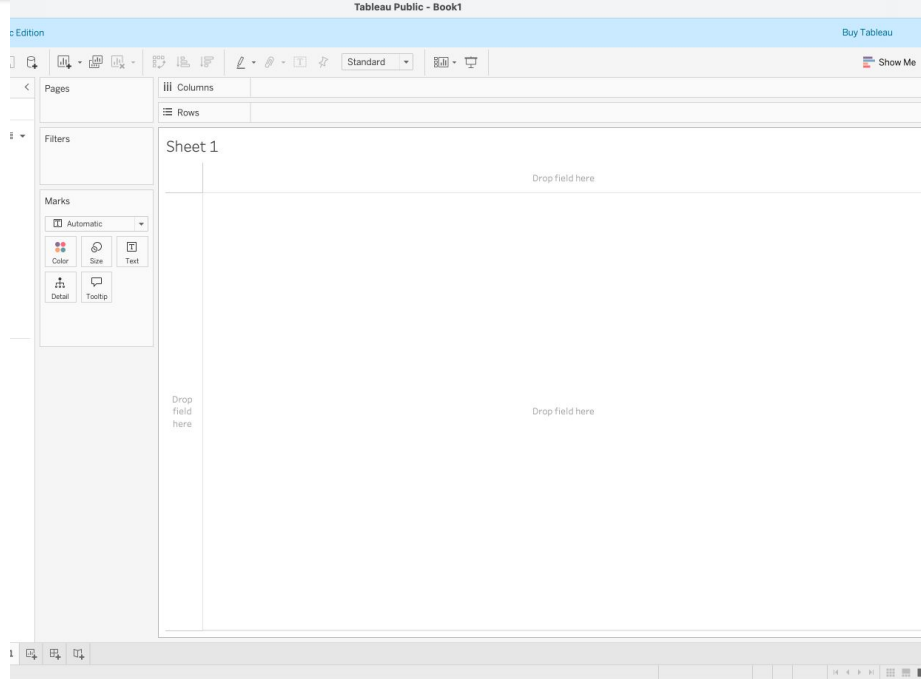
DSCI 454
Week 3, Lecture 1



What are we learning this lecture?

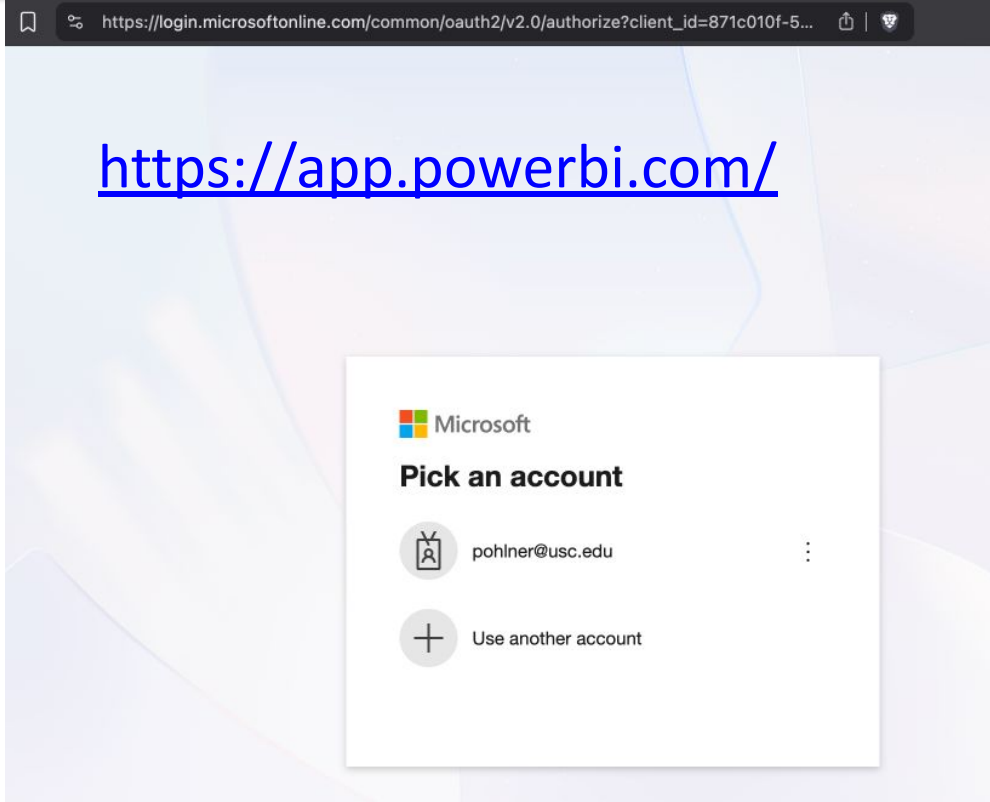
- No Code Visualization Tools
 - Power BI (PBI) Introduction
- In-Class Demo / Student Follow Along

No Code Visualization Tools in General



Most follow a standard templated layout with a central WYSIWYG canvas and data, widget, and filter type palette(s) on the right or left hand pane you can click and drag from onto the canvas

Power BI (PBI) Login



Make sure you login
with your USC email
to activate the Univ.'s
Institutional /
Organizational
provisions / licenses

PBI Start / Home Page



Power BI Home

Search

New report Click here to create new

New items saved to: My workspace

Recommended

My workspace
You frequently open this

Sales Analytics
Aamina Thasneem Khaleel featured this

Sales Analytics
Popular in your org

Open

Open

Open

Expl Gettr

Saved models and reports shown here

Recent Favorites My apps From external orgs

Filter by keyword Filter

Name	Type	Opened	Location	Endorsement	Sensitivity
Netflix_Report_v2	Report	31 seconds ago	My workspace	—	—
Netflix_Model_v2	Semantic model	an hour ago	My workspace	—	—
My workspace	Workspace	an hour ago	Workspaces	—	—
FinanceReportSaveFile	Report	4 hours ago	My workspace	—	—
FinanceReportSaveFile	Report	5 hours ago	My workspace	—	—

The Start / Home page allows you to create a new model or report or open previously saved reports models or reports

PBI Import Data



Power BI My workspace

Build your first report

- Add and prepare your data
- Generate a premade report
- Customize to suit your needs

Add data to start building a report

Get Data OneLake catalog Excel CSV Paste or manually enter data

Import CSV or Excel

Pick a published semantic model

Don't see the source you're looking for? [Download the desktop app.](#)

Looking for the old experience? [Excel](#) or [CSV](#)

- For this course / semester, we will be starting with mostly CSVs and XLSs from sites like Kaggle et al. but PBI supports many others incl.
 - built-in samples,
 - data lakes,
 - OneDrive,
 - and more / still growing

PBI Import CSV



Power Query

Get data

Connect to data source

Text/CSV
File
[Learn more](#)

Connection settings

☐ Link to file ☒ Upload file

Drag a file here to upload
or

Connection credentials

Connection
Create new connection

Connection name
Connection

Data gateway
(none)

Authentication kind
Organizational account

You are currently signed in as:

 **Gregory Pohlner**
pohlner@usc.edu
[Switch account](#)

Privacy Level
None

☐ This connection can be used with on-premises data gateways and VNet data gateways.

Cancel Next

- For this course / semester, we will be starting with mostly CSVs and XLSs
- All example data files used in lecture will usually be hosted on Bright Space
- For HW assignments, students will be asked to find their own data sets from sites like Kaggle et al.

PBI Preview Data



Power Query

Get data

Preview file data

URL: https://uscedu-my.sharepoint.com/personal/pohlner_usc_edu/Documents/Apps/Microsoft Power Query/Uploaded Files/netflix1.2.csv

File origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data type detection: Based on first 200 rows

show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min	Documenta
s3	TV Show	Ganglands	Julien Leclercq	France	9/24/2021	2021	TV-MA	1 Season	Crime TV S
s6	TV Show	Midnight Mass	Mike Flanagan	United States	9/24/2021	2021	TV-MA	1 Season	TV Dramas,
s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	9/22/2021	2021	TV-PG	91 min	Children &
s8	Movie	Sankofa	Haile Gerima	United States	9/24/2021	1993	TV-MA	125 min	Dramas, Inc
s9	TV Show	The Great British Baking Show	Andy Devonshire	United Kingd...	9/24/2021	2021	TV-14	9 Seasons	British TV S
s10	Movie	The Starling	Theodore Melfi	United States	9/24/2021	2021	PG-13	104 min	Comedies,
s939	Movie	Motu Patlu in the Game of Zones	Suhas Kadav	India	5/1/2021	2019	TV-Y7	87 min	Children &
s13	Movie	Je Suis Karl	Christian Schwchow	Germany	9/23/2021	2021	TV-MA	127 min	Dramas, Int
s940	Movie	Motu Patlu in Wonderland	Suhas Kadav	India	5/1/2021	2013	TV-Y7	76 min	Children &
s941	Movie	Motu Patlu: Deep Sea Adventure	Suhas Kadav	India	5/1/2021	2014	TV-Y7	76 min	Children &
s942	Movie	Motu Patlu: Mission Moon	Suhas Kadav	India	5/1/2021	2013	TV-Y7	71 min	Children &
s852	Movie	99 Songs (Tamil)	Not Given	Pakistan	5/21/2021	2021	TV-14	131 min	Dramas, Int
s471	Movie	Bridgerton - The Afterparty	Krysia Plonka, Kristian Mercado	United States	7/13/2021	2021	TV-14	39 min	Movies
s730	Movie	Bling Empire - The Afterparty	Krysia Plonka, Kristian Mercado	United States	6/12/2021	2021	TV-MA	36 min	Movies
s731	Movie	Cobra Kai - The Afterparty	Krysia Plonka, Kristian Mercado	United States	6/12/2021	2021	TV-MA	34 min	Movies
s913	Movie	The Circle - The Afterparty	Krysia Plonka, Kristian Mercado	United States	5/7/2021	2021	TV-14	35 min	Comedies
s4	TV Show	Jailbirds New Orleans	Not Given	Pakistan	9/24/2021	2021	TV-MA	1 Season	Docuseries
s15	TV Show	Crime Stories: India Detectives	Not Given	Pakistan	9/22/2021	2021	TV-MA	1 Season	British TV S
s3232	Movie	True: Winter Wishes	Mark Thornton, Todd Kauffman	United States	11/26/2019	2019	TV-Y	46 min	Children &
s4832	TV Show	True: Magical Friends	Mark Thornton, Todd Kauffman	United States	6/15/2018	2018	TV-Y	1 Season	Kids' TV
s4833	TV Show	True: Wonderful Wishes	Mark Thornton, Todd Kauffman	United States	6/15/2018	2018	TV-Y	1 Season	Kids' TV
s4857	TV Show	Dance & Sing with True	Mark Thornton, Todd Kauffman	United States	5/18/2018	2018	TV-Y	1 Season	Kids' TV
s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Not Given	9/24/2021	2021	PG	91 min	Children &
s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Not Given	9/23/2021	2021	TV-MA	1 Season	Crime TV S
s17	Movie	Europe's Most Dangerous Man: Otto Skorzeny in Spain	Pedro de Echave García, Pablo Azorin Williams	Not Given	9/22/2021	2020	TV-MA	67 min	Documenta
s7930	Movie	Samudri Lootere	Anirban Majumder	Not Given	6/18/2019	2018	TV-Y	65 min	Children &

Back Cancel Next

- You usually can accept the default selections for:
 - Origin / Encoding
 - Delimiter
 - Comma for CSV
 - Tab for TSV
 - Data type detection
 - More later
- Click green “Next” button in bottom right corner

PBI Power Query / Data Transform



Power Query

Search (Alt + Q)

PBI tries to do the initial data type conversions automatically

Home Transform Add column View Help

Queries [1]

netflix1 2

```
Table.TransformColumnTypes(#"Promoted headers", {"show_id", type text}, {"type", type text}, {"title", type text}, {"director", type text}, {"country", type text}, {"date_added", type date}, {"release_year", Int64.Type}, {"rating", type text}, {"duration", type text}, {"listed_in", type text})
```

show_id	type	title	director
s1	Movie	Dick Johnson Is Dead	Kirsten Johnson
s3	TV Show	Ganglands	Julien Leclercq
s6	TV Show	Midnight Mass	Mike Flanagan
s14	Movie	Confessions of an Invisible Girl	Bruno Garotti
s8	Movie	Sankofa	Haile Gerima
s9	TV Show	The Great British Baking Show	Andy Devonshire
s10	Movie	The Starling	Theodore Melfi
s939	Movie	Motu Patlu in the Game of Zones	Suhas Kadav
s13	Movie	Je Suis Karl	Christian Schwochow
s940	Movie	Motu Patlu in Wonderland	Suhas Kadav
s941	Movie	Motu Patlu: Deep Sea Adventure	Suhas Kadav
s942	Movie	Motu Patlu: Mission Moon	Suhas Kadav
s852	Movie	99 Songs (Tamil)	Not Given
s471	Movie	Bridgerton - The Afterparty	Krysia Plonka, Kristian Mercado
s730	Movie	Bling Empire - The Afterparty	Krysia Plonka, Kristian Mercado
s731	Movie	Cobra Kai - The Afterparty	Krysia Plonka, Kristian Mercado
s913	Movie	The Circle - The Afterparty	Krysia Plonka, Kristian Mercado
s4	TV Show	Jailbirds New Orleans	Not Given
s15	TV Show	Crime Stories: India Detectives	Not Given
s3232	Movie	True: Winter Wishes	Mark Thornton, Todd Kauffman
s4832	TV Show	True: Magical Friends	Mark Thornton, Todd Kauffman
s4833	TV Show	True: Wonderful Wishes	Mark Thornton, Todd Kauffman
s4857	TV Show	Dance & Sing with True	Mark Thornton, Todd Kauffman
s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha
s12	TV Show	Bangkok Breaking	Kongkiat Komesiri
s17	Movie	Europe's Most Dangerous Man: Otto Skorzeny in Spain	Pedro de Echave García, Pablo Azorín Williams

Query settings

Properties

Name: netflix1 2

Applied steps

Source

Promoted ...

Changed c...

Change History

Completed (1.20 s) Columns: 10 Rows: 99+

Create a report Cancel

- PBI tries to do the initial data type conversions automatically
 - More later
- There a reversible change history pane on the right hand side

PBI Power Query / Data Transform Before



Power Query

Search (Alt + Q)

Home Transform Add column View Help

Queries [1] netflix1 1

```
Table.TransformColumnTypes(#"Promoted headers", {"show_id", type text}, {"type", type text}, {"title", type text}, {"director", type text}, {"country", type text}, {"date_added", type date}, {"release_year", Int64.Type}, {"rating", type text}, {"duration", type text}, {"listed_in", type text}})
```

country	date_added	release_year	rating	duration	listed_in
States	9/25/2021	2020	PG-13	90 min	Documentaries
	9/24/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Action &
States	9/24/2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
	9/22/2021	2021	TV-PG	91 min	Children & Family Movies, Comedies
States	9/24/2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies
Kingd...	9/24/2021	2021	TV-14	9 Seasons	British TV Shows, Reality TV
States	9/24/2021	2021	PG-13	104 min	Comedies, Dramas
	5/1/2021	2019	TV-Y7	87 min	Children & Family Movies, Comedies, Music & Musicals
ony	9/23/2021	2021	TV-MA	127 min	Dramas, International Movies
	5/1/2021	2013	TV-Y7	76 min	Children & Family Movies, Music & Musicals
	5/1/2021	2014	TV-Y7	76 min	Children & Family Movies, Comedies
	5/1/2021	2013	TV-Y7	71 min	Children & Family Movies, Comedies
an	5/21/2021	2021	TV-14	131 min	Dramas, International Movies, Music & Musicals
States	7/13/2021	2021	TV-14	39 min	Movies
States	6/12/2021	2021	TV-MA	36 min	Movies
States	6/12/2021	2021	TV-MA	34 min	Movies
States	5/7/2021	2021	TV-14	35 min	Comedies
an	9/24/2021	2021	TV-MA	1 Season	Docuseries, Reality TV
an	9/22/2021	2021	TV-MA	1 Season	British TV Shows, Crime TV Shows, Docuseries
States	11/26/2019	2019	TV-Y	46 min	Children & Family Movies
States	6/15/2018	2018	TV-Y	1 Season	Kids' TV
States	6/15/2018	2018	TV-Y	1 Season	Kids' TV
States	5/18/2018	2018	TV-Y	1 Season	Kids' TV
ven	9/24/2021	2021	PG	91 min	Children & Family Movies
ven	9/23/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Action &
ven	9/22/2021	2020	TV-MA	67 min	Documentaries, International Movies

Query settings

Properties

Name netflix1 1

Applied steps

- Source
- Promoted ...
- Changed c...
- Replaced v...
- Replaced v...
- Replaced v...
- Renamed c...
- Renamed C...
- Renamed C...
- Renamed C...
- Renamed C...
- Renamed C...
- Renamed C...
- Renamed C...
- Renamed C...

Completed (0.91 s) Columns: 10 Rows: 99+

Cancel Save

- The Netflix data set's "Duration" column is problematic because it contains mixed data type text e.g.
 - 1 Season
 - 2 Seasons
 - 93 minutes
- We need to convert this to a number

PBI Power Query / Data Transform Replace Values



Power Query

Search (Alt + Q)

Home Transform Add column View Help

Queries [1] netflix1 1

```
Table.TransformColumnTypes(#"Promoted headers", {"show_id", type text}, {"type", type text}, {"title", type text}, {"director", type text}, {"country", type text}, {"date_added", type date}, {"release_year", Int64.Type}, {"rating", type text}, {"duration", type text}, {"listed_in", type text})
```

1. Right click column header

2. Select

country	date_added	release_year	rating	duration	listed_in
States	9/25/2021				Documentaries
	9/24/2021				Crime TV Shows, International TV Shows, TV Action &
States	9/24/2021				TV Dramas, TV Horror, TV Mysteries
	9/22/2021				Children & Family Movies, Comedies
States	9/24/2021				Dramas, Independent Movies, International Movies
Kingd...	9/24/2021				British TV Shows, Reality TV
States	9/24/2021				Comedies, Dramas
	5/1/2021				Children & Family Movies, Comedies, Music & Musical
any	9/23/2021				Dramas, International Movies
	5/1/2021				Children & Family Movies, Music & Musicals
	5/1/2021				Children & Family Movies, Comedies
	5/1/2021				Children & Family Movies, Comedies
an	5/21/2021				Dramas, International Movies, Music & Musicals
States	7/13/2021				Movies
States	6/12/2021				Movies
States	6/12/2021				Movies
States	5/7/2021				Comedies
an	9/24/2021				Docuseries, Reality TV
an	9/22/2021				British TV Shows, Crime TV Shows, Docuseries
States	11/26/2019				Children & Family Movies
States	6/15/2018				Kids' TV
States	6/15/2018				Kids' TV
States	5/18/2018				Kids' TV
ven	9/24/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Action &
ven	9/23/2021	2020	TV-MA	67 min	Documentaries, International Movies
ven	9/22/2021				

Completed (0.91 s) Columns: 10 Rows: 99+

- One option is to right click the column header and choose “Replace Values”
- Replace “Seasons” with nothing / blank
- Replace “Season” with nothing / blank
- Replace “min” with nothing / blank
- Note order and leading space

PBI Power Query / Data Transform Replace Values 2



country	date_added	release_year	rating	duration
United States	9/25/2021	2020	PG-13	90
France	9/24/2021	2021	TV-MA	1 Season

Replace values ?

Replace one value with another in the selected columns.

☒ Basic ☐ Advanced

Note leading space

Value to find

Seasons

Replace with

Replace with nothing / leave blank to delete

Cancel

OK

- Replace “Seasons” with nothing / blank
- Replace “Season” with nothing / blank
- Replace “min” with nothing / blank
- Note order and leading space

PBI Data Transform Change Type Custom Column



Power Query

Home Transform **Add column** View Help

Column from examples

Queries [1] netflix1 1

Table.ReplaceValue("#"Changed column type", " min", "", Replacer.ReplaceText, {"duration"})

Custom column ?

Add a column that is computed from other columns or values.

New column name * durationNum

Custom column formula * = Text.Select([duration], {"0".. "9", ".", "-"})

Data type Decimal number

Available column(s) director, country, date_added, release_year, rating, duration, listed_in

Insert column

Cancel OK

Learn more about Power Query formulas

21	s4832	TV Show	True: Magical Friends	Mark Thornton, Todd Kauffman
22	s4833	TV Show	True: Wonderful Wishes	Mark Thornton, Todd Kauffman
23	s4857	TV Show	Dance & Sing with True	Mark Thornton, Todd Kauffman
24	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha
25	s12	TV Show	Bangkok Breaking	Kongkiat Komesiri
26	s17	Movie	Europe's Most Dangerous Man: Otto Skorzeny in Spain	Pedro de Echave Garcia, Pablo Azorin Williams

Columns: 10 Rows: 99+

1. From Transform Data / Power Query view, Click "Add Column"
2. Click "Custom Column"
3. Set new column name
4. Set new column data type
5. Paste expression
 - a. 0-9 numbers, commas, and dashes only
6. Click green "OK" button in bottom right corner

PBI Power Query / Data Transform Change Type



Power Query

Search (Alt + Q)

Home Transform Add column View Help

Queries [1] netflix1

Table.ReplaceValue("#"Replaced value 1", " Season", "", Replacer.ReplaceText, {"duration"})

actor	country	date_added	release_year	rating	duration
Johnson, Ryan	United States	9/25/2021			
Leclercq, Jean	France	9/24/2021			
Flanagan, John	United States	9/24/2021			
Garotti, Paulo	Brazil	9/22/2021			
Gerima, Haile Gerima	United States	9/24/2021			
Devonshire, The	United Kingdom	9/24/2021			
Iorio Melfi, The	United States	9/24/2021			
Kadav, The	India	5/1/2021			
Schwochow, The	Germany	9/23/2021			
Kadav, The	India	5/1/2021			
Kadav, The	India	5/1/2021			
iven, The	Pakistan	5/2/2021			
Plonka, Kristian Mercado	United States	7/13/2021			
Plonka, Kristian Mercado	United States	6/12/2021			
Plonka, Kristian Mercado	United States	6/12/2021			
Plonka, Kristian Mercado	United States	5/7/2021			
iven, The	Pakistan	9/24/2021			
iven, The	Pakistan	9/22/2021			
Thornton, Todd Kauffman	United States	11/26/2019			
Thornton, Todd Kauffman	United States	6/15/2018			
Thornton, Todd Kauffman	United States	6/15/2018			
Thornton, Todd Kauffman	United States	5/18/2018			
Cullen, José Luis Ucha	Not Given	9/24/2021			
Komesiri, The	Not Given	9/23/2021	2021	TV-MA	1
de Echave García, Pablo Azorín Williams	Not Given	9/22/2021	2020	TV-MA	67

Columns: 10 Rows: 99+

1 Right-click column header

2

3

Query settings

Properties

Name netflix1

Applied steps

Source

Promoted ...

Changed c...

Replaced v...

Replaced v...

Replaced v...

Changed c...

Renamed C...

Renamed C...

Decimal number

Currency

Whole number

Percentage

Date

Date/Time

Date/Time/Zone

Time

Duration

Text

True/False

Binary

Using locale...

Cancel Save

- Right-click a text type column header
- Select “Change type”
- Choose new type
 - In this case “Whole Number”
 - Other options like “Decimal”, etc.
- Rinse and repeat for all text to numeric columns
- Click green “Save” button in bottom right corner

PBI Semantic Model View



The screenshot shows the Power BI Semantic Model View interface for a model named 'Netflix_Model v2'. The interface is divided into several panes: a left sidebar with navigation options (Home, Copilot, Create, Browse, OneLake catalog, Apps, Scorecards, Workspaces, My workspace, Netflix_Model v2, IMDbModel), a top ribbon with tabs (File, Home, Help) and a '0. Select Confirm in "Editing" mode' button, a main area with a '1. Note / Confirm Semantic Model converted / updated successfully.' message, a 'Properties' pane for the selected 'netflix1' table, and a 'Data' pane showing the table's structure. The 'Data' pane lists columns: Country, Date Added, Director, Duration, Genre, Rating, Release Year, show_id, Title, and Type. The 'Duration' column is highlighted, and its properties are shown in the 'Properties' pane, including Name, Description, Display folder, Is hidden, Data type (Whole number), Format (Whole number), and Percentage format. A pull-down menu for the 'Duration' column is open, showing various data type icons (sigma for numeric, calendar for date/time). Annotations with red arrows point to specific elements: '5. File -> New Report' points to the File tab; '4. Select to return to Power Query view' points to the 'Transform Refresh data' button; '0. Select Confirm in "Editing" mode' points to the 'Editing' dropdown; '1. Note / Confirm Semantic Model converted / updated successfully.' points to the success message; '2. Note / Confirm icon symbols match desired type (Calendar icon for date (/ time))' points to the calendar icon in the pull-down menu; '2. Note / Confirm icon symbols match desired type (Sigma symbol for numeric types)' points to the sigma icon in the pull-down menu; '3. Can also try to "auto detect" and change data type in Model View via this pull down but may have to return to "Transform Data" / "Power Query" view for more complex transformations. See Step 4.' points to the pull-down menu.

5. File -> New Report

4. Select to return to Power Query view

0. Select Confirm in "Editing" mode

1. Note / Confirm Semantic Model converted / updated successfully.

2. Note / Confirm icon symbols match desired type (Calendar icon for date (/ time))

2. Note / Confirm icon symbols match desired type (Sigma symbol for numeric types)

3. Can also try to "auto detect" and change data type in Model View via this pull down but may have to return to "Transform Data" / "Power Query" view for more complex transformations. See Step 4.

0. Select / Confirm in edit mode
1. Confirm Data Transforms completed successfully w/o error
2. Confirm updated data type icons / symbols
3. Alternative / Quick / Auto Convert Data Type / Format
4. Return to Transform Data view when needed
5. File -> "Create New Report" when ready

PBI: Report



The screenshot shows the Microsoft Power BI interface. The top navigation bar includes 'Power BI My workspace' and a search bar. The left sidebar contains navigation icons for Home, Copilot, Create, Browse, OneLake catalog, Apps, Scorecards, Monitor, Workspaces, My workspace, and Untitled report. The main canvas area displays a 'Build visuals with your data' prompt with a red callout: 'Blank WYSIWYG canvas Drag data columns (from Data palette) on right here'. The right-hand pane is divided into three sections: 'Filters' (with 'Filters on this page' and 'Filters on all pages' sections), 'Visualizations' (with a 'Build visual' dropdown and various chart icons), and 'Data' (showing a list of columns from a 'netflix1' dataset). A red callout points to the 'Data' pane with the text: '1. Drag y-axis measurement / column to canvas first Start w/ the "Title" column'. The bottom status bar shows 'Page 1' and a '+' icon.

Start by dragging your y-axis measurement / column from the right hand Data palette to the WYSIWYG canvas on the left

PBI: Report: Table View



- After the drag n drop, a single text type column will result in a data view
- Now drag n drop your x-axis over
- Let's try Year Added

PBI: Report: Table View w/ Year



1. Expand Date Hierarchy,
2. Select Year,
3. Drag to Title Table on left

PBI: Report: Blank Line?



The screenshot shows the Microsoft Power BI interface. On the left is the navigation pane with icons for Home, Copilot, Create, Browse, OneLake catalog, Apps, Scorecards, Monitor, Workspaces, My workspace, and an Untitled report. The main area is a blank report canvas. On the right are the Filters, Visualizations, and Data panes. The Visualizations pane shows various chart types, with the line chart icon highlighted. The Data pane shows a table named 'netflix1' with columns like Country, Date Added, Date Hierarc..., Year, Quarter, Month, Day, Director, Duration, Genre, Rating, Release Year, show_id, Title, and Type. The Title column is selected. The X-axis and Y-axis sections in the Visualizations pane are empty. The Filters pane shows 'Date Added - Year is (All)' and 'Title is (All)'. The bottom status bar shows 'Page 1' and a '+' icon.

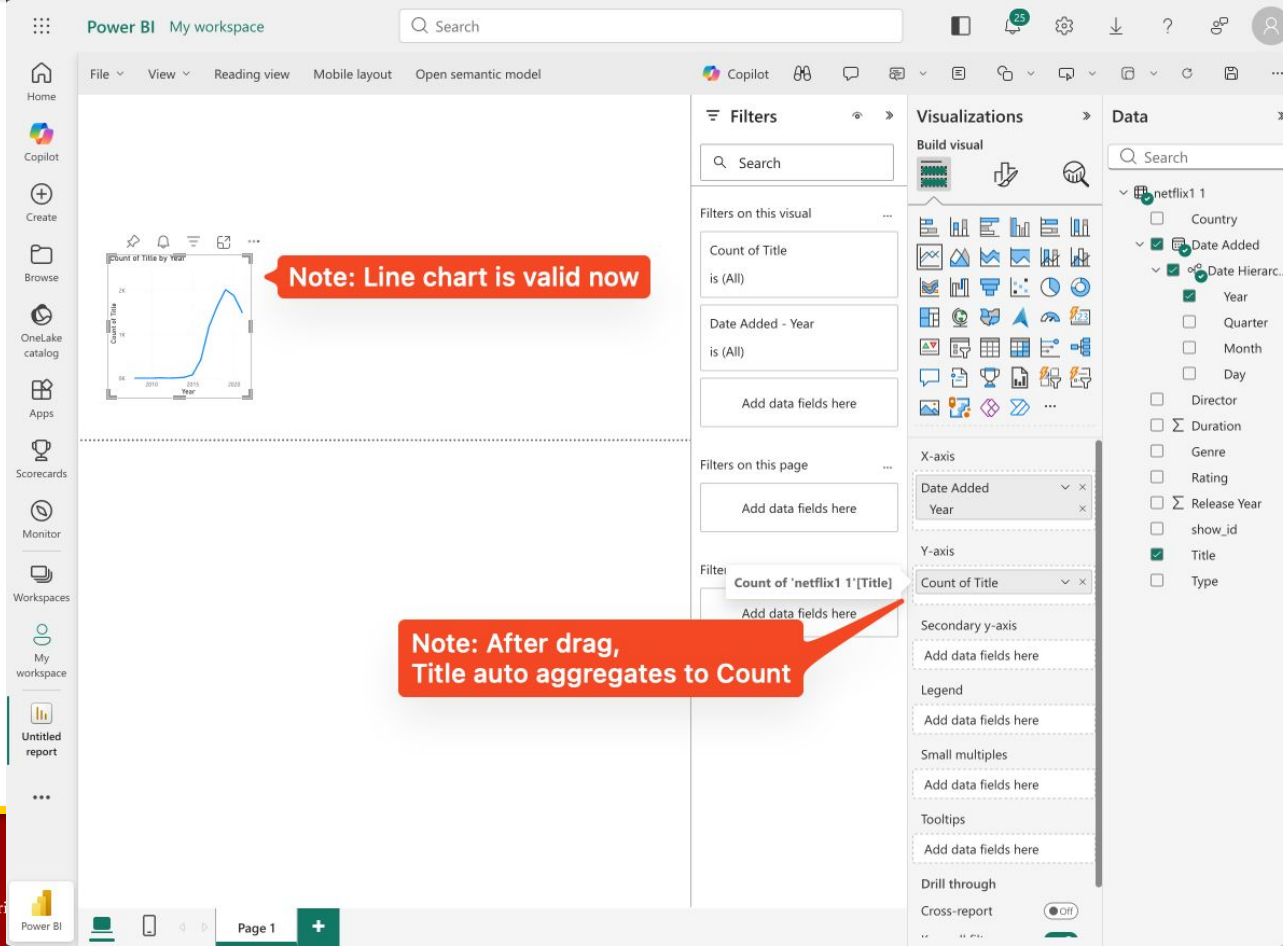
1. Select chart widget on canvas

2. Select Line type

3. Note it's blank because of both on X-axis

4. Drag Title to Y-Axis

PBI: Report: Line Fixed



PBI: Report: Line Add Type Legend



Power BI My workspace

3. Note Line chart updates to show a line for Movies and TV Shows w/ a Legend

Count of Title by Year and Type

Type: Movies, TV Shows

Count of Title

Year

Filters

Search

Filters on this visual

Count of Title is (All)

Date Added - Year is (All)

Type is (All)

Add data fields here

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Visualizations

Build visual

X-axis: Date Added, Year

Y-axis: Count of Title

Secondary y-axis: Add data fields here

Legend: Type

Small multiples: Add data fields here

Tooltips: Add data fields here

Drill through: Add data fields here

Cross-report: Off

Keep all filters: On

Data

Search

netflix1

- Country
- Date Added
- Date Hierarc...
- Year
- Quarter
- Month
- Day
- Director
- Duration
- Genre
- Rating
- Release Year
- show_id
- Title
- Type

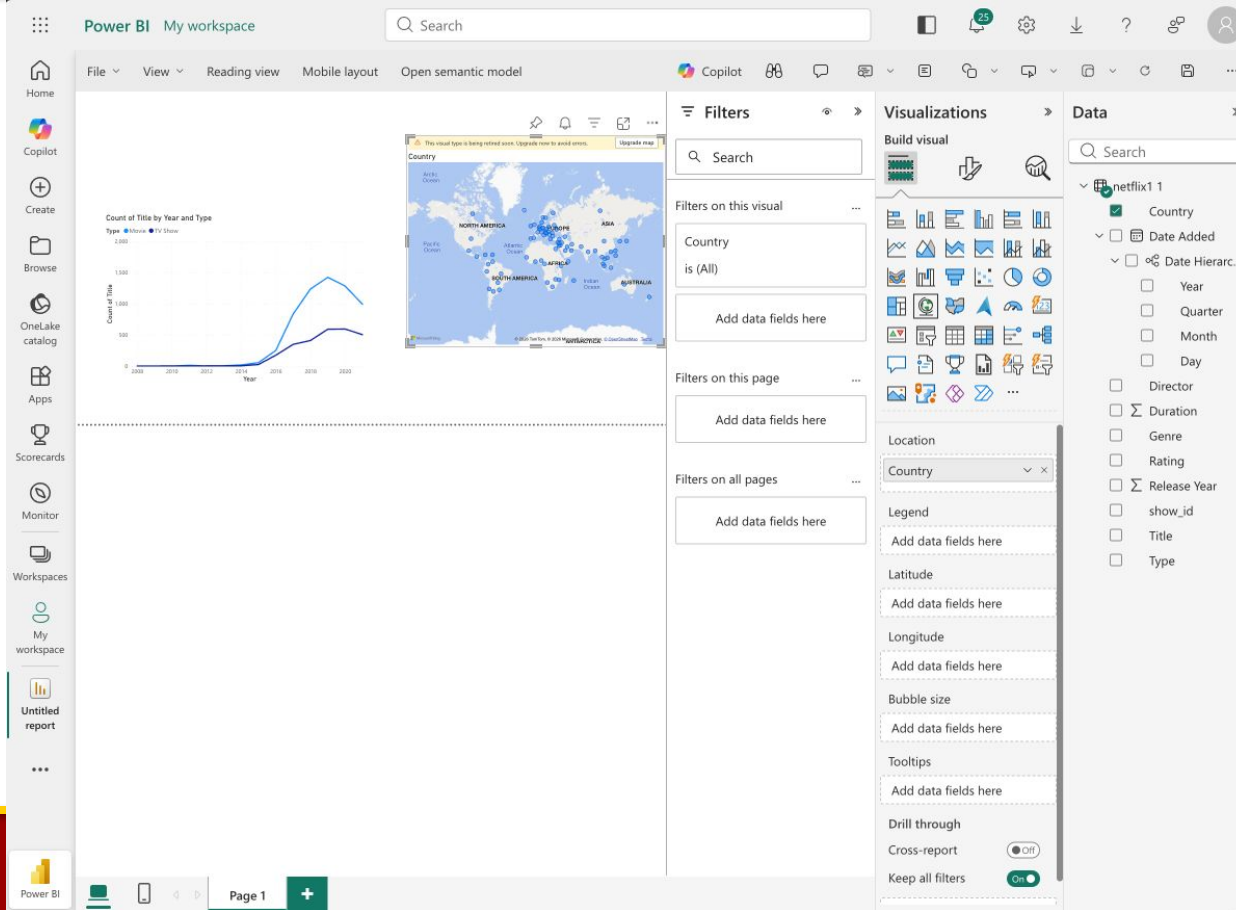
1. Drag Type

2. To Legend

USC Viterbi School of Engineering

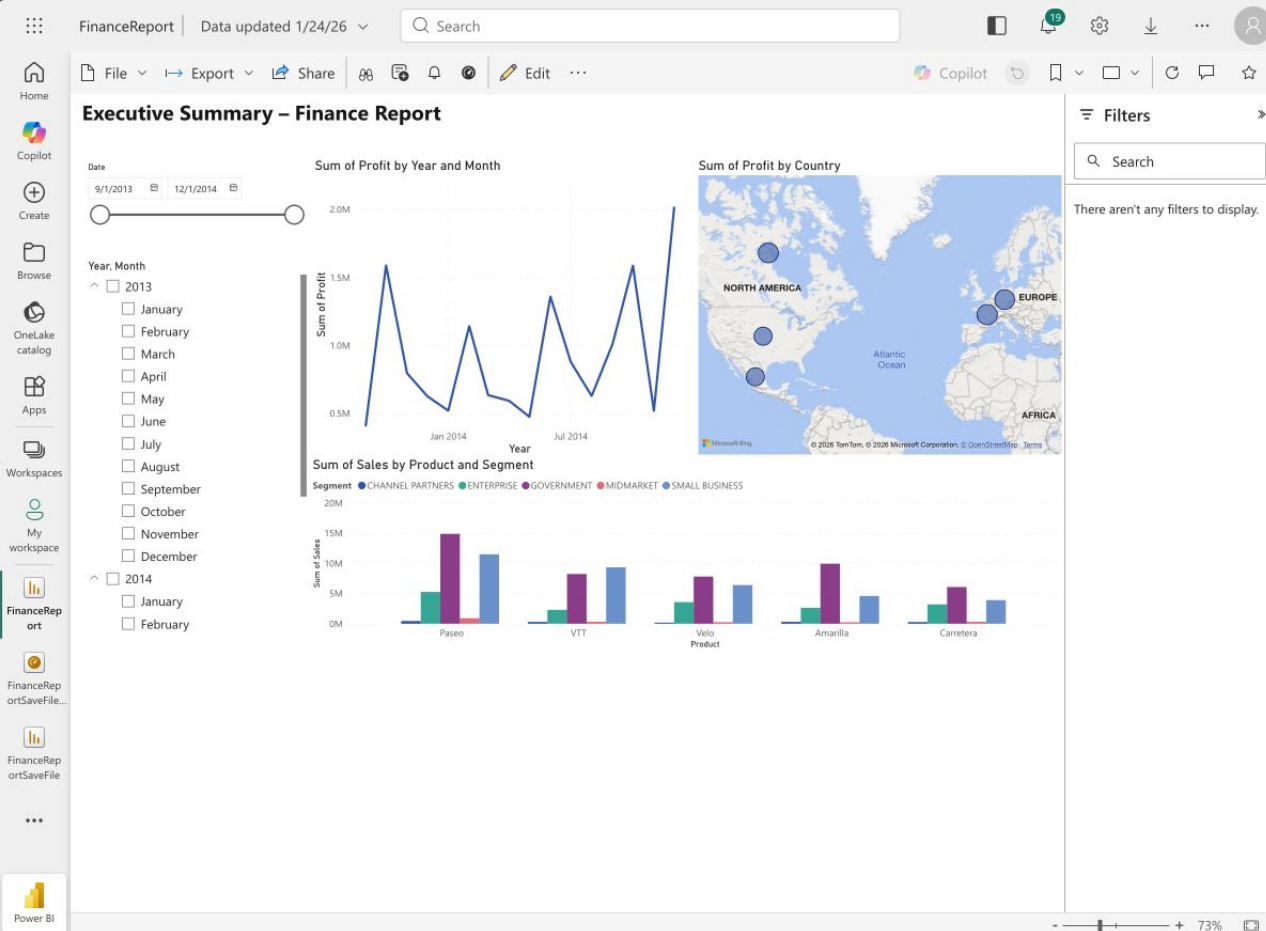
Page 1

PBI: Report: Geographic



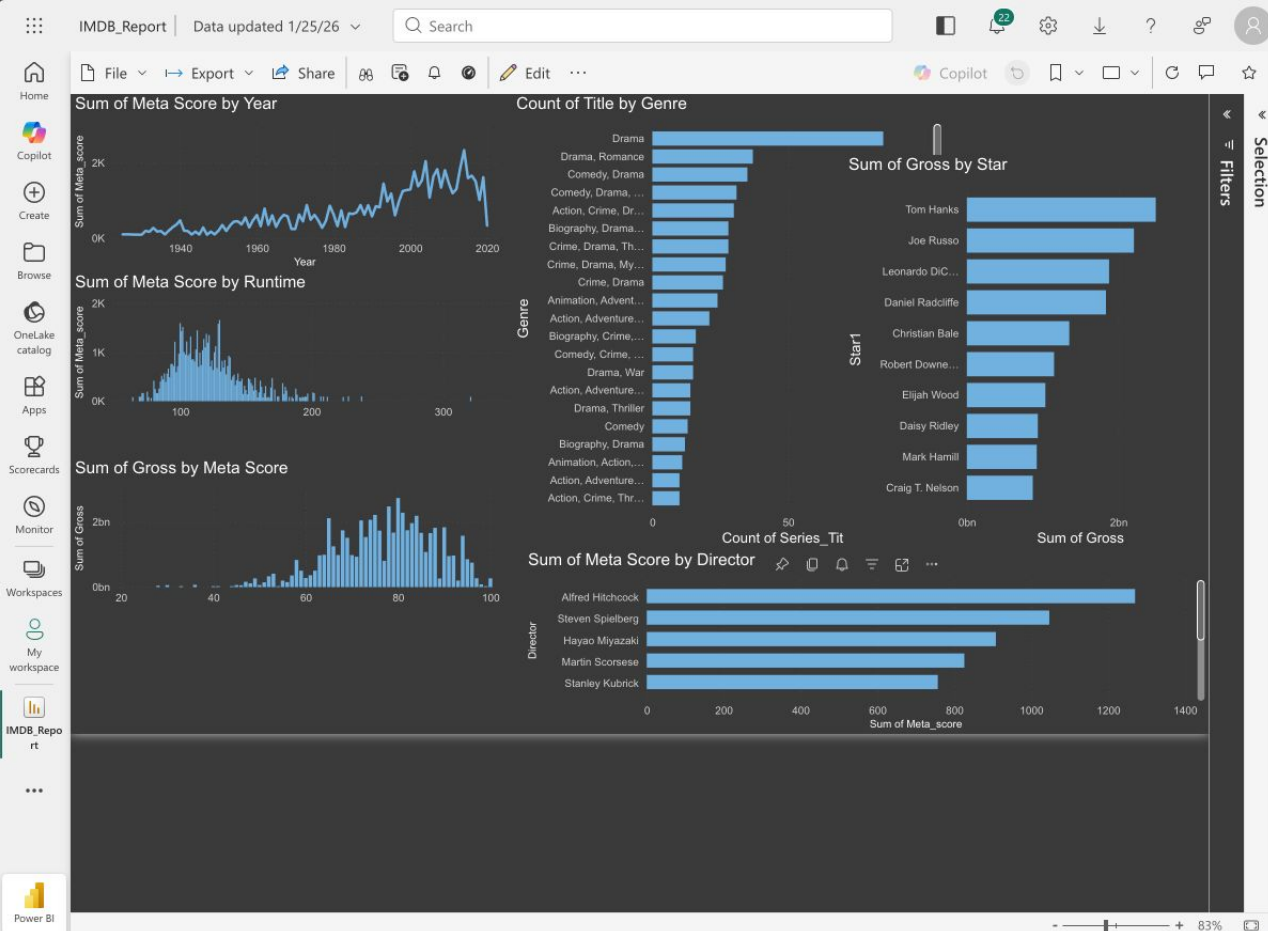
- PBI will auto detect countries, regions, cities, etc. in text columns
- Just drag and drop over
- Click “Upgrade Map” to upgrade to Azure maps button if prompted

PBI Microsoft (MSFT) Tutorial Finance Report



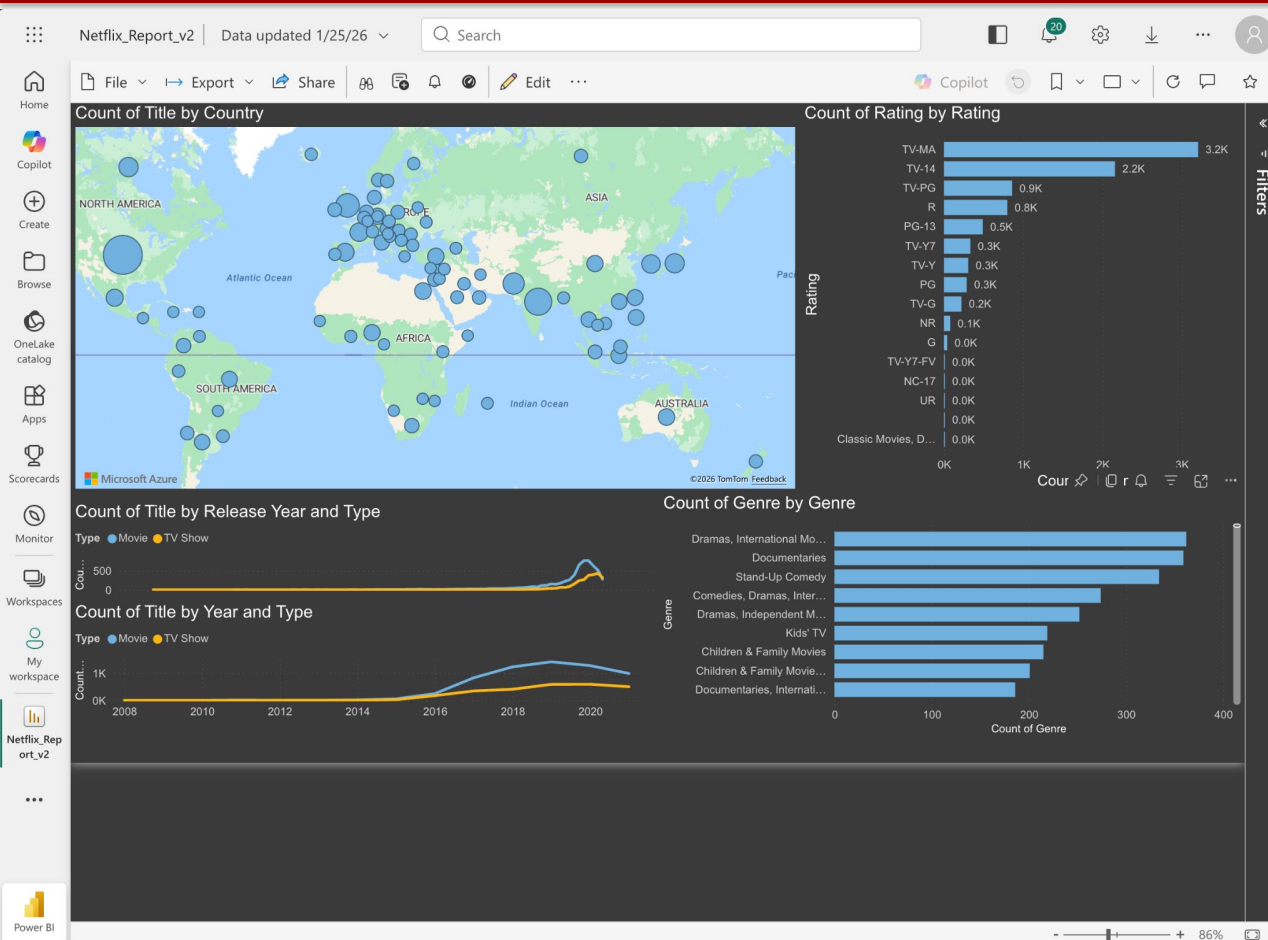
- MSFT's Tutorial using sample finance data
- [MSFT Power BI Tutorial](#)
 - [MSFT Financial Sample Data Set](#)

PBI IMDB Report



- [Kaggle IMDB Data Set](#)

PBI Netflix Report



- [Kaggle Netflix Data Set](#)



Power BI Misc. Specifics

- Histogram is called “Clustered Column Chart”
- Simple bar count charts (like in Netflix ex.) can be made by dragging the same column for both X and Y-axis
- View -> Themes
 - Aesthetics only, be cognizant of accessibility
- View -> Selection Pane for order,
 - e.g. PowerPoint’s “Bring to Front” or “Send to Back”
- Extract numbers only from a mixed text column
 - Transform Data -> New Custom Column -> Power Query:
 - `NewColumnName = Text.Select([YourColumnName], {"0".."9", ".", "-"})`



Resources and References

- Power BI Login: <https://app.powerbi.com/>
- [MSFT Power BI Tutorial](#)
 - [MSFT Financial Sample Data Set](#)
 - [More MSFT Sample Data](#)
- [Kaggle IMDB Data Set](#)
- [Kaggle Netflix Data Set](#)

Questions?

